Performance Evaluation and System Description

1. Performance Evaluation

The proposed Deepfake Audio Detection System was evaluated using a held-out test dataset consisting of both real and AI-generated audio samples. The final model selected for deployment was an optimized XGBoost classifier trained on handcrafted audio features extracted from speech recordings.

Test Set Results

Metric	Value
Accuracy	98.28%
Precision	98%
Recall	99%
F1-Score	98%
Error Rate	1.79%

Classification Report

Class	Precision	Recall	F1-Score	Support
Real (0)	0.99	0.98	0.98	1396
Fake (1)	0.98	0.99	0.98	1396
Macro Avg	0.98	0.98	0.98	2792
Weighted Avg	0.98	0.98	0.98	2792

Confusion Matrix

```
[[1364   32]
 [  16 1380]]
```

Where:

- True Positives (TP): 1380
- True Negatives (TN): 1364
- False Positives (FP): 32
- False Negatives (FN): 16

The confusion matrix demonstrates that the model correctly classified the vast majority of both real and fake audio samples while maintaining a very low number of misclassifications.

Equal Error Rate (EER)

Equal Error Rate (EER) is an important metric in biometric and audio verification systems. It represents the point where the False Acceptance Rate (FAR) equals the False Rejection Rate (FRR).

The EER was computed using the model's probability outputs and ROC analysis.

Equal Error Rate (EER): 1.79%

A lower EER indicates better discrimination capability between real and deepfake audio samples.

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2. Data Preprocessing

The audio dataset contained both genuine human speech recordings and AI-generated speech samples.

Preprocessing Steps

1. Audio files were loaded using the Librosa library.
2. All recordings were sampled at 16 kHz.
3. Audio duration was standardized to 2 seconds.
4. Audio signals were converted to mono format.
5. Feature vectors were extracted from each audio sample.
6. Missing values were checked and handled appropriately.
7. Feature scaling was performed using StandardScaler.
8. The dataset was split into training and testing subsets for model evaluation.

Preprocessing ensured consistency across all audio samples and improved model learning.

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3. Feature Extraction

Feature extraction is a crucial stage in audio classification systems because machine learning models cannot directly process raw waveform data efficiently.

The system extracts multiple spectral and temporal characteristics from each audio recording.

3.1 Mel Frequency Cepstral Coefficients (MFCC)

MFCCs capture the timbral characteristics of speech and are widely used in speech recognition and speaker verification systems.

Extracted Features:

- MFCC Mean (13 coefficients)
- MFCC Standard Deviation (13 coefficients)

Total MFCC Features:

```
26 Features
```

3.2 Zero Crossing Rate (ZCR)

Zero Crossing Rate measures how frequently the signal crosses the zero amplitude axis.

Extracted Features:

- Mean ZCR
- Standard Deviation ZCR

Total:

```
2 Features
```

3.3 Spectral Centroid

Spectral Centroid indicates the center of mass of the frequency spectrum and provides information about signal brightness.

Extracted Features:

- Mean Spectral Centroid
- Standard Deviation Spectral Centroid

Total:

```
2 Features
```

3.4 Spectral Bandwidth

Spectral Bandwidth measures the spread of frequencies around the spectral centroid.

Extracted Features:

- Mean Spectral Bandwidth
- Standard Deviation Spectral Bandwidth

Total:

```
2 Features
```

3.5 Chroma Features

Chroma features represent the distribution of spectral energy among the twelve pitch classes.

Extracted Features:

- Chroma Mean (6 selected features)
- Chroma Standard Deviation (6 selected features)

Total:

```
12 Features
```

Final Feature Vector

The complete feature vector consists of:

```
MFCC Features                 : 26
Zero Crossing Rate Features   : 2
Spectral Centroid Features    : 2
Spectral Bandwidth Features   : 2
Chroma Features               : 12
----------------------------------
Total Features                : 44
```

These features provide a compact representation of audio characteristics that help distinguish real speech from synthesized speech.

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4. Model Architecture

XGBoost Classifier

The final classification model used in this project is XGBoost (Extreme Gradient Boosting), a powerful ensemble learning algorithm based on gradient-boosted decision trees.

XGBoost was selected because of its:

- High classification accuracy
- Fast training speed
- Strong generalization capability
- Ability to handle nonlinear relationships
- Built-in regularization mechanisms

Model Configuration

```
XGBClassifier(
    n_estimators=300,
```

```
    max_depth=8,
    learning_rate=0.05,
    subsample=0.8,
    colsample_bytree=0.8,
    random_state=42,
    eval_metric="logloss"
)
```

Hyperparameter Description

Parameter	Value	Description
n_estimators	300	Number of decision trees
max_depth	8	Maximum depth of each tree
learning_rate	0.05	Step size used during boosting
subsample	0.8	Fraction of training samples used per tree
colsample_bytree	0.8	Fraction of features used per tree
random_state	42	Reproducibility seed
eval_metric	logloss	Optimization objective

Prediction Pipeline

The complete prediction workflow is shown below:

```
Audio File
     │
     ▼
Feature Extraction
     │
     ▼
44-Dimensional Feature Vector
     │
     ▼
StandardScaler
     │
     ▼
XGBoost Classifier
     │
     ▼
Probability Scores
     │
     ▼
Real / Fake Prediction
```

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5. Conclusion

The proposed Deepfake Audio Detection System successfully identifies AI-generated speech using handcrafted audio features and an optimized XGBoost classifier. The system achieved an accuracy of 98.28% with strong precision, recall, and F1-score values, demonstrating its effectiveness in distinguishing between authentic and synthetic audio recordings.

The low error rate and expected low Equal Error Rate further indicate the model's robustness and suitability for practical deepfake audio detection applications.